PBL Secondary Research

Systolic Array Matrix Multiplication

Team No:1

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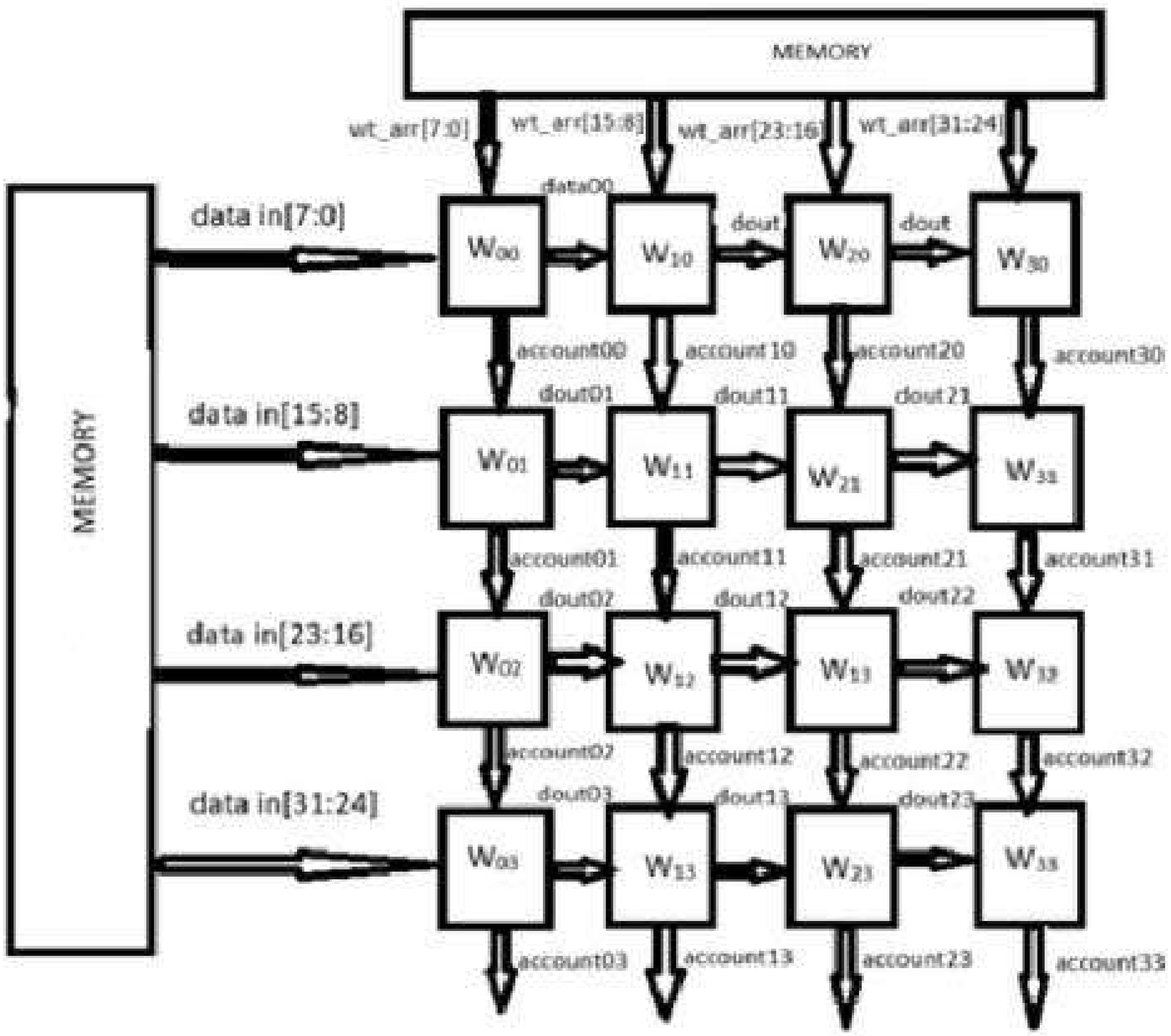
Problem Statement:

Matrix-matrix multiplication is a fundamental operation in scientific computing, digital signal processing, and machine learning. However, traditional sequential algorithms for multiplying large matrices suffer from high computational complexity and memory bottleneck, making them inefficient for large-scale data processing. As datasets grow, there is a pressing need for parallel hardware architectures that can perform matrix operations more efficiently.

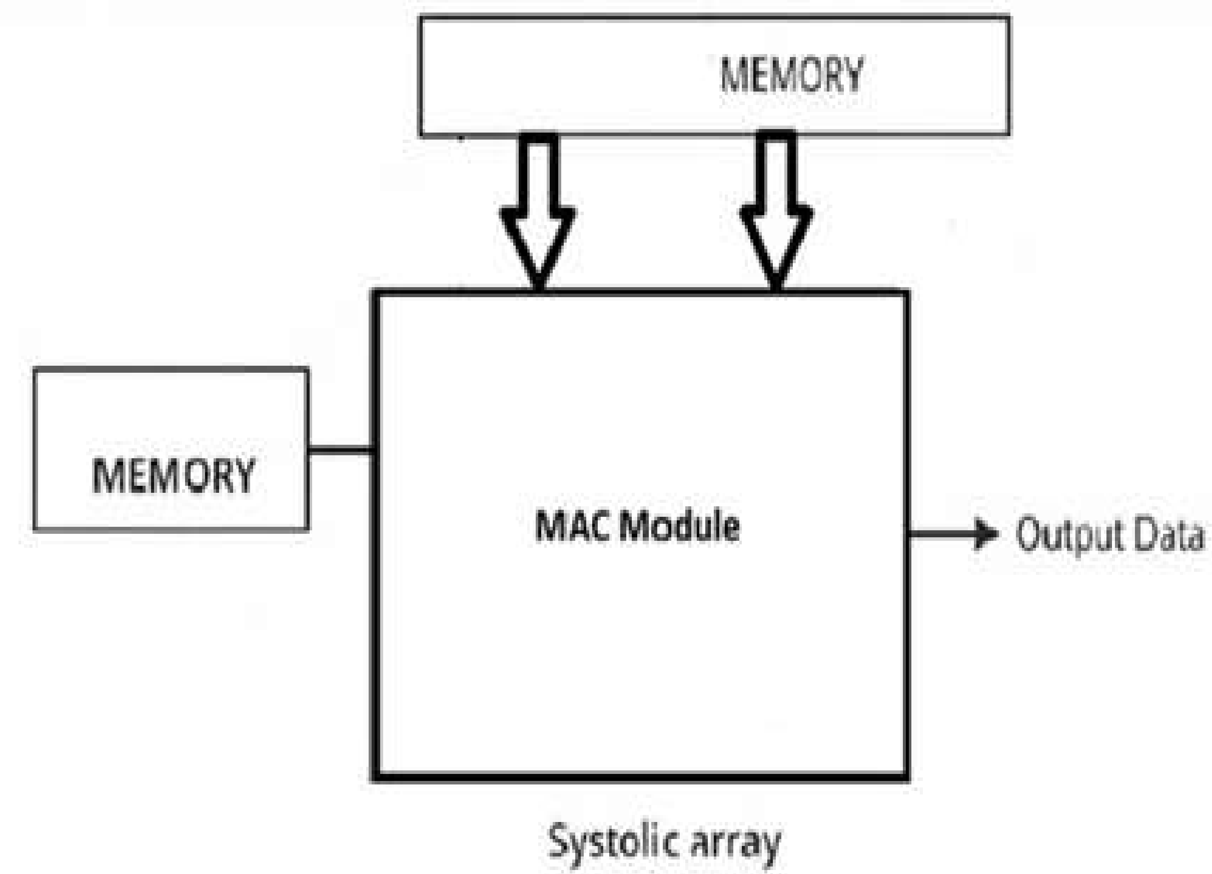
Objectives:

- ❑ **Reduce computation time :** Design a systolic array that performs matrix multiplication faster than traditional sequential algorithms.
- ❑ **Enable parallelism** :Use multiple processing elements working simultaneously to handle large-scale matrix operations efficiently.
- ❑ **Scalability** :Ensure the architecture can be extended to support larger matrices without significant performance loss.
- ❑ **Hardware suitability:** Implement the design in FPGA kit.

Architecture Diagram

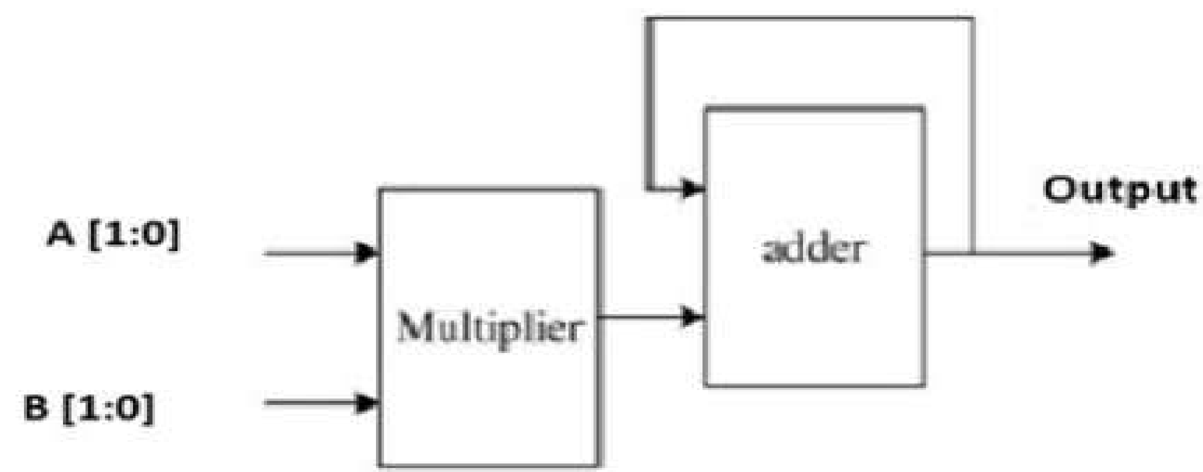


Block Diagram



Leaf Cell Design

MAC Module



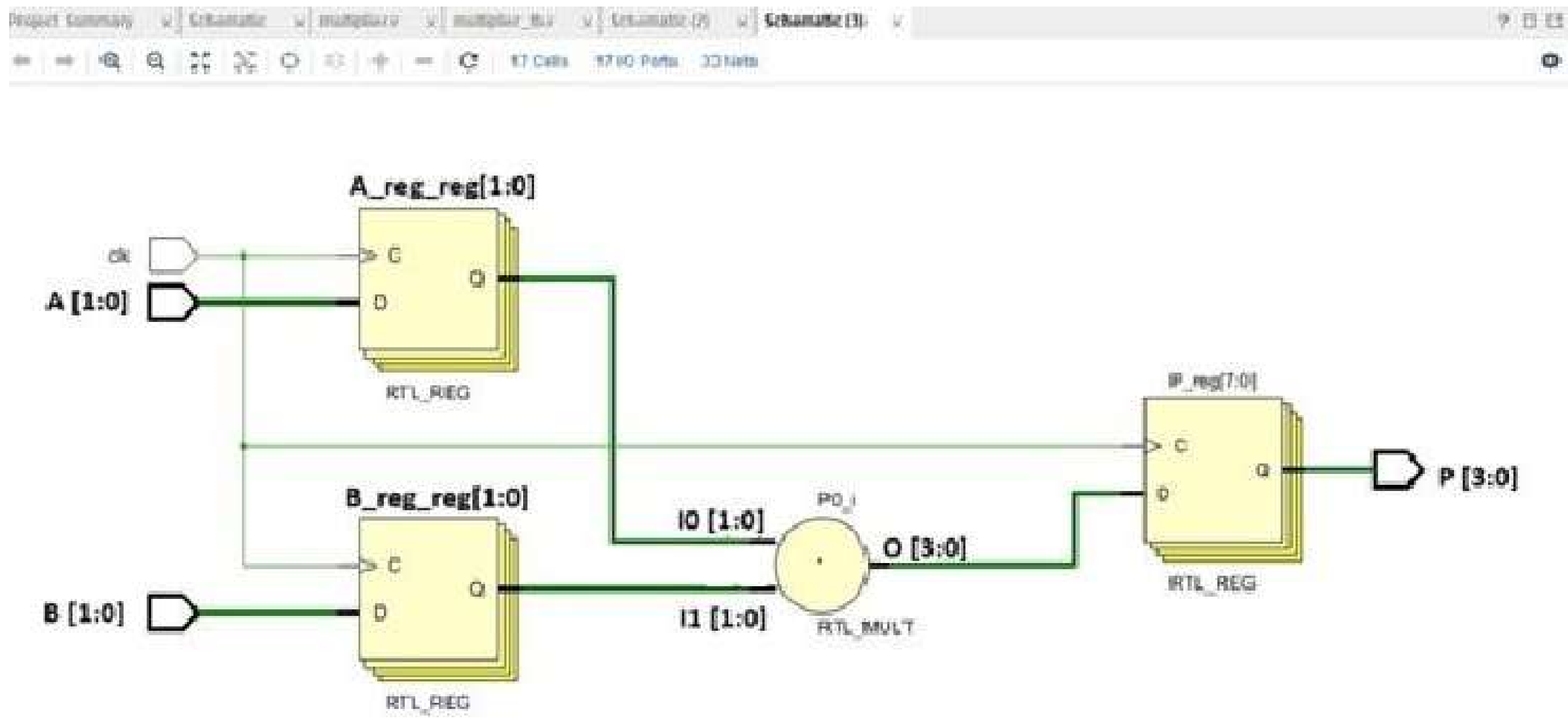
Leaf Cells:

- Multiplier
- Adder

❑ A MAC module, or Multiply–Accumulate module, is a small hardware unit designed to perform two operations in sequence: multiplication and addition.

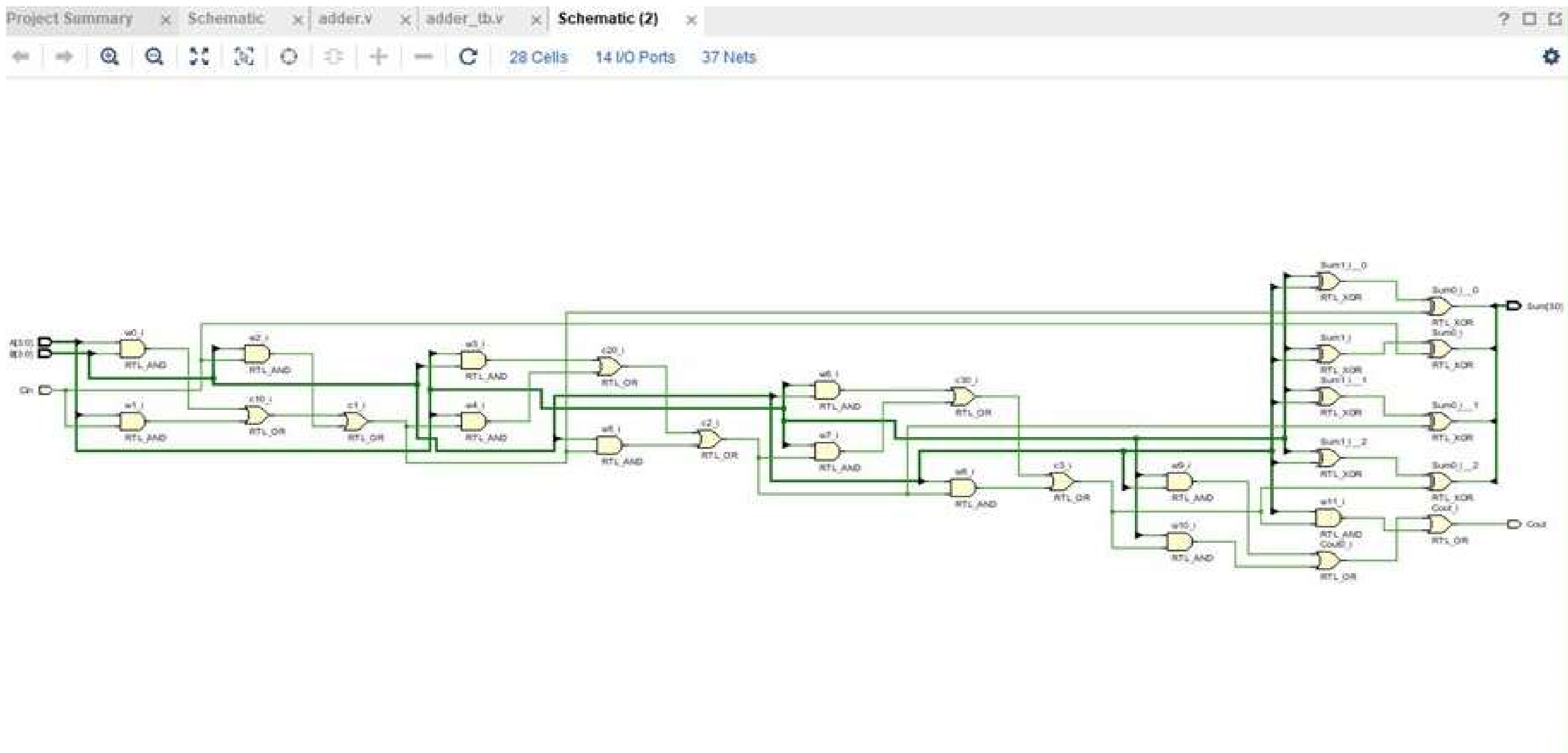
$$\mathrm{Sum}_{i,j}(t+1) = \mathrm{Sum}_{i,j}(t) + (A_{i,k} \times B_{k,j})$$

Multiplier



- A multiplier is a digital circuit or arithmetic unit that takes two input numbers and produces their product as the output.
- Here we are using 2 bit inputs (A[1:0],B[1:0])

Adder



- ❑ An adder is a basic digital circuit that performs the arithmetic operation of addition. It takes two binary numbers as inputs and produces their sum as the output, often along with a carry signal if the result exceeds the bit capacity.
- ❑ Here Cin is taken as optional.

PBL Work Distribution

13- march 2026 – Problem statement selection and research on existing papers and solutions

1-April 2026 - listing of rules and regulations within the team
2-April 2026 – 4th April 2026 – research paper submission by each team member to compare the difference between the existing solutions
20 –April 2026 – 2-April 2026 - worked on Assignment 1

Contribution of Each team members in assignment -1

Armaan (4SO24EC012) - Problem statement elaboration and objectives
Simulation of Adder (Leaf cell)
Work assigned on: 20-04-2026
Submitted on: 01-05-2026

Abhilash M (4SO24EC002) - Block diagram and architectural diagram
Simulation of Multiplier (Leaf cell)
Work assigned on: 20-04-2026
Submitted on: 01-05-2026

Ashish D’souza (4SO24EC014)- research gap identification and literature review
Simulation of Mac module (leaf cell)
Work assigned on: 20-04-2026
Submitted on: 01-05-2026

Avinia Viola Pinto (4SO24EC020)- Comparative analysis, formula and explanation for each leaf cell, report documentation
Work assigned on: 20-04-2026
Submitted on: 01-05-2026

https://github.com/Avinia08/Group-1-PBL.git